

Tutorial No - 4

Q.1] List and explain tools and techniques for quality control

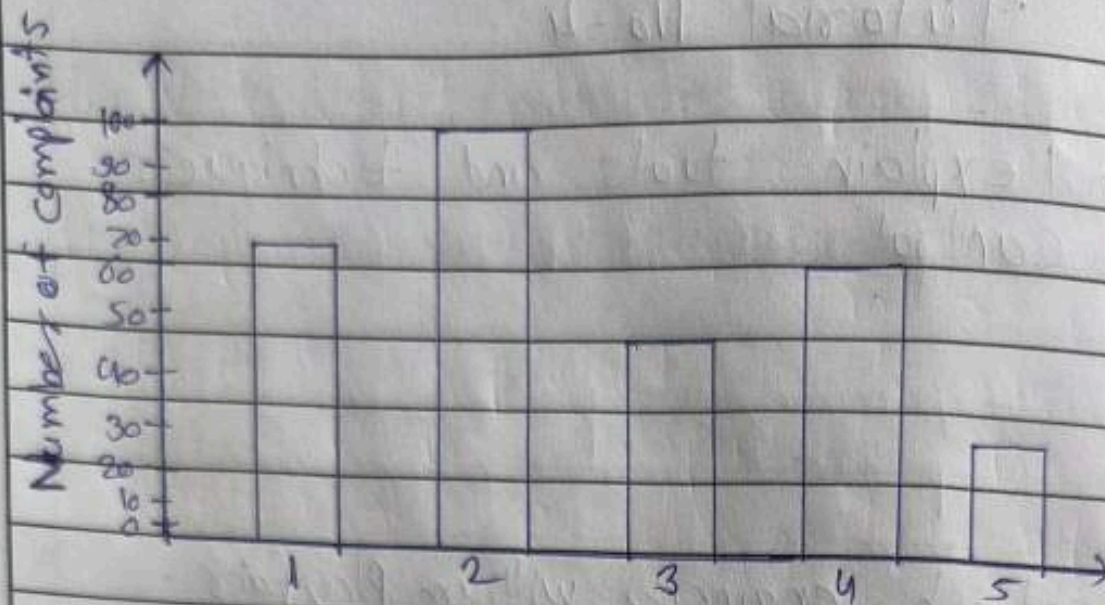
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① Chart -

- In 1786, Scottish engineer William Playfair introduced various types of charts, including pie charts and bar charts.
- Chart defined as graphical representation of data.
- The chart is typically linked with worksheet.
- In quality management, visual representation simplifies complex data.

② Histogram -

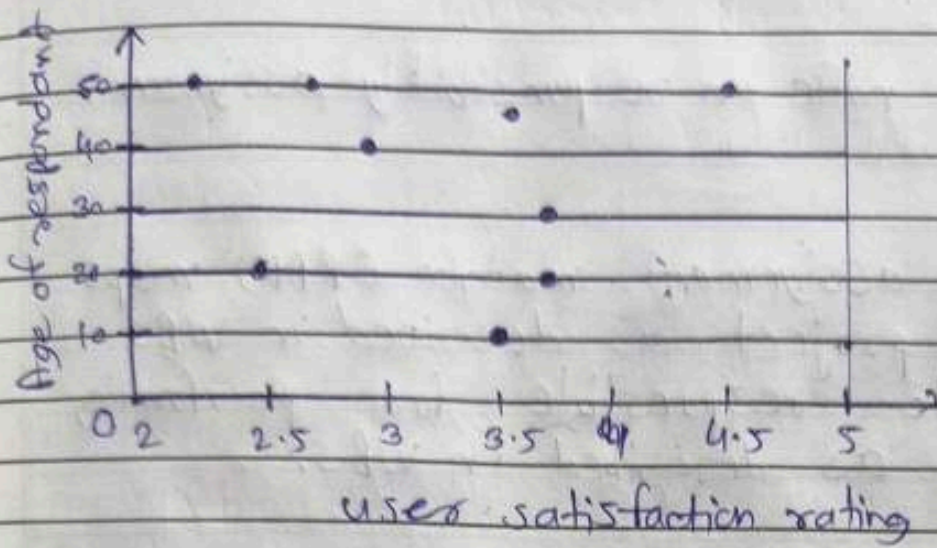
- Karl Pearson introduced the histogram in 1891.
- A histogram is one of quality management tools in project management whose structures are similar to bar graphs.



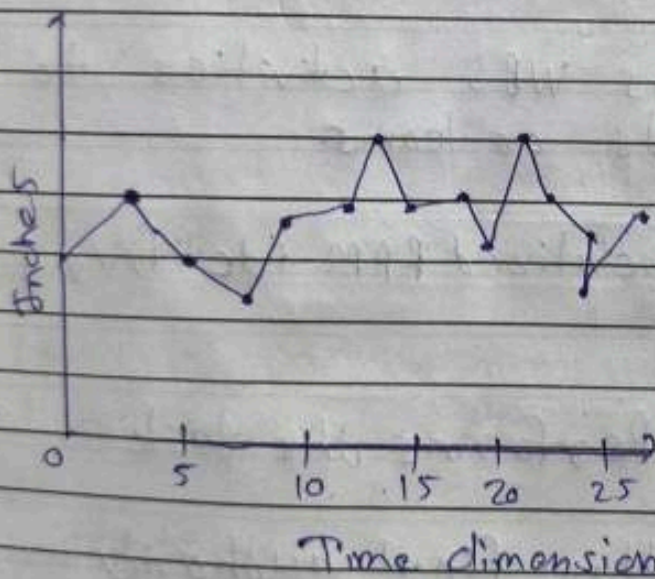
week
sample histogram

③ Scatter Diagram:-

- A scatter diagram also called scatterplot, scatter chart or scatter graph, was by John F.W. Herschel in 1833.
- The graph uses dots to display the values of 2 different numeric variables.
- The numerical data is paired from one variable on each axis.
- In quality management, the scatter diagram helps identify relationships between variables to understand the quality defects possible causes.



④ Control Chart -



- The control chart or Stewart control charts is a tool that was introduced by Walter.
- It is graphical tool used to study the process of change over time.
- In quality management the graph is widely used by professionals to time and money.

Q.2] Write short note on responsibility assignment matrices?

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- A responsibility assignment matrix (RAM) maps the work of project, as described in WBS to the people responsible for performing the work as described in WBS.

- RAM help clarify who is responsible for what especially useful in both small and large project.

① Basic RAM - Assigns WBS activities to responsible individuals or teams.

② RACI Chart - A specialized RAM identifying four roles -

① R (Responsible) - Performs the task

② A (Accountable) - Has final authority or approval

③ C (Consulted) - Provides necessary information

④ I (Informed) - Needs updates on task progress

- These tools enhance communication and coordination in project management.

WBS activities →

obs
units

	1.1.1	1.1.2	1.1.3	1.1.4	1.1.5	1.1.6	1.1.7	1.1.8
System engineering	R	RP						
Software development			RP					
Hardware development				PR				
Test engineering	P							
Quality Assurance					RP			
Configuration mana.						RP		
Integrated logi. supp							P	
Training								RP

R = responsible organizational unit

P = performing organizational unit

Q.3] Write short note on RACI chart?

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- A RACI chart is a project management tool used to clarify roles and responsibilities for tasks within a project
- It stands for

① R - Responsible - A person who does the work

② A - Accountable - the person who owns the task and makes final decisions

③ C - Consulted - people who provide input and expertise.

④ I-informed- People who need updates on progress or results

- The chart lists tasks vertically and team members or role horizontally with each cell marked by R, A, C or I
- Each task should have only one accountable person to avoid confusion
- RACI charts improve clarity reduce overlap and ensure everyone understands their role in the project

Sample RACI chart

	Car owner	shop owner	mechanic	Parts
pay Parts and services	A, R			
determine parts and services needed	C		A, R	C
supply parts		C		
Install parts	I	A	R	A, R

Q.4] Explain Tackman model of team development.

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- Stages of Model-

Forming

This initial stage involves team members getting to know each other, establishing roles and defining the team's purpose and goals.

Storming

As the team works together, conflicts and disagreements may arise as members try to position and establish their roles, potentially leading to tension and friction.

Norming

The team begins to establish norms, expectations and working styles, resolving conflicts and building trust and cohesion.

Performing

The team functions at the highest level with members working collaboratively and efficiently towards common goals.

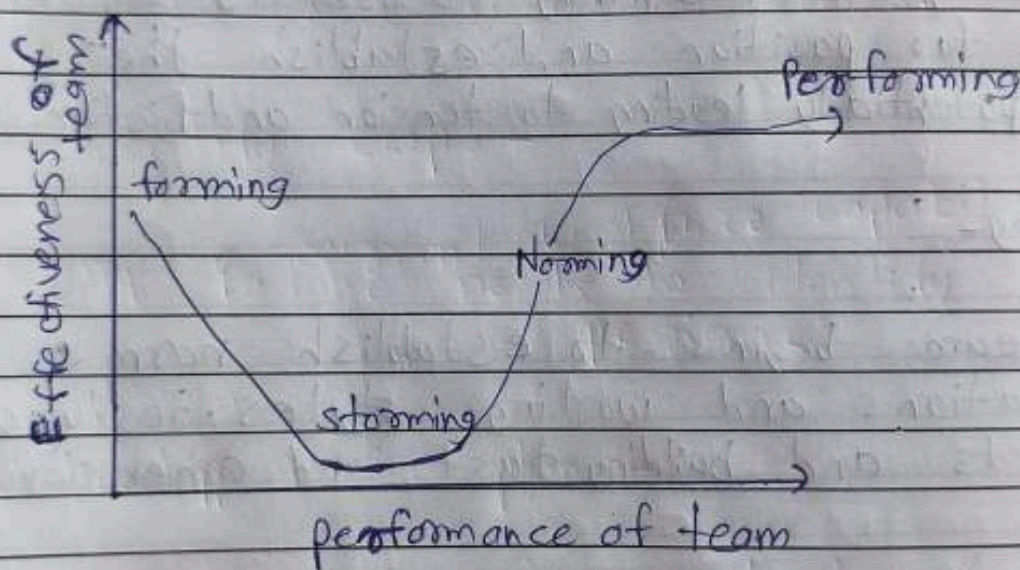
Adjourning

This is the final stage, which occurs when the team's project or task is completed and the team disbands or transitions to a new phase.

Key takeaways -

- ① understanding the stages
- ② Predictable progression
- ③ leadership role
- ④ Adaptability

Tuckman's Team and Group development Model.



Q.5] Explain the process "Developing a Human Resource plan" in detail.

- The development of a resource management plan involves identifying and documenting resources, roles responsibilities skills and documenting for a project
- This plan is typically divided into a human resource management plan and plan.

The human resources plan includes-

- ① An organization chart
- ② Detailed roles and responsibilities
- ③ A staffing management plan
- ④ Optionally a team charter to guide team operations.

Key takeaways-

- Human resource planning is strategy companies use to maintain a steady supply of skilled employees while avoiding staffing shortage or surplus.
- HRP need to meet short-term staffing challenges while adapting to changing conditions in business environment over the long term.
- Having good HRP strategy in place mean productivity and profitability for company.

* Steps of Human resources planning-

- ① Analyze objectives- clearly define the organization's strategic objectives.

② Assess current workforce -

- Conduct a thorough analysis of the existing workforce, including skills, experience and performance.

③ Forecast future needs -

- Predict future HR requirements based on business goals and industry trends.

④ Identify Gaps -

- Compare current workforce capabilities with future needs to identify any gaps in skills, experience.

⑤ Develop Action plans -

- Create strategies and plans to address the identified gaps, which may include recruitment, training, development.

⑥ Implement the plan -

- Put the action plans into practice, which may involve hiring, training or other HR initiatives.

⑦ Monitor and evaluate -

- Continuously monitor the implementation of the HR plans.